

P10. Real-World Evaluation of Two Cooperative Intersection Management Approaches

中英双语博士精读笔记

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Theme / 主题: Real-world cooperative intersection management

Method family / 方法族: Real-world cooperative maneuver planning evaluation / 真实道路协同机动规划评估

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1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Real-world cooperative intersection management”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Real-world cooperative intersection management. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Focuses on cooperative maneuver planning evaluation, not a JSSP formulation.

1.2 核心科学问题 / Core Scientific Question

协同交叉口管理方法在混合交通仿真与真实原型车测试中是否仍能减少停车和通行时间？

Do cooperative intersection-management approaches reduce stops and crossing time in both mixed-traffic simulation and real prototype drives?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Compares multi-scenario prediction and graph-based reinforcement learning approaches in mixed-traffic simulation and real drives.

- 安全/避碰贡献 (safety contribution): Evaluates criticality metrics while operating with prototype connected automated vehicles in public traffic.
- 平滑/停止避免贡献 (smoothing contribution): Reports substantial reductions in crossing time and number of stops for cooperative maneuver planning.
- 创新力度评判 (reviewer judgement): 强验证价值: 真实道路评估稀缺, 是论文实验设计和指标选择的优质参考。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Real-world cooperative maneuver planning evaluation	真实道路协同机动规划评估	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: 评估指标聚合 / Evaluation-metric aggregation, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

This paper has been accepted for publication in the IEEE Intelligent Transportation Systems Magazine (ITSM). This is the accepted version of the paper, which has not been fully edited and the layout may differ from the original publication. A CC BY 4.0 version is available from <https://doi.org/10.18725/OPARU-59080>

Real-World Evaluation of two Cooperative Intersection Management Approaches

Marvin Klimke^{*,1,2}, Max Bastian Mertens^{*,1}, Benjamin Völz², and Michael Buchholz¹

Abstract—Cooperative maneuver planning promises to significantly improve traffic efficiency at unsignalized intersections by leveraging connected automated vehicles. Previous works on this topic have been mostly developed for completely automated traffic in a simple simulated environment. In contrast, our previously introduced planning approaches are specifically designed to handle real-world mixed traffic. The two methods are based on multi-scenario prediction and graph-based reinforcement learning, respectively. This is the first study to perform evaluations in a novel mixed traffic simulation framework as well as real-world drives with prototype connected automated vehicles in public traffic. The simulation features the same connected automated driving software stack as deployed on one of the automated vehicles. Our quantitative evaluations show that cooperative maneuver planning achieves a substantial reduction in crossing times and the number of stops. In a realistic environment with few automated vehicles, there are noticeable efficiency gains with only slightly increasing criticality metrics.

Index Terms—Connected automated driving, cooperative planning, driver model, scenario prediction, optimization, reinforcement learning, graph neural network, mixed traffic, simulation, real-world evaluation, traffic efficiency.

I. INTRODUCTION

URBAN traffic is prone to inefficiencies and disturbances due to the ever-increasing volume of traffic [16]. This manifests, e.g., at smaller intersections, where static priority rules are the prevailing method for coordination of vehicles. Connected automated driving opens up new opportunities to improve urban traffic efficiency by leveraging communication links between vehicles and possibly infrastructure systems. Moreover, edge computing resources are becoming available in urban areas that enable, e.g., populating and maintaining of a collective environment model (EM) of an area of interest. This setup allows connected automated vehicles (CAVs) to improve their local planning algorithms by incorporating data from the server-side EM, as it was shown in [4] at a suburban three-way intersection in Ulm-Lehr, Germany.

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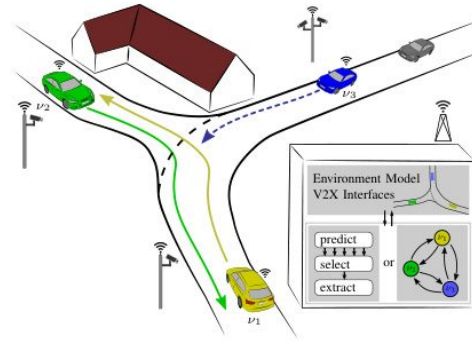


Fig. 1. Cooperative maneuver at an unsignalized intersection. The CAV v_3 on the major road gives way to the turning CAV v_2 and the CAV v_1 on the subordinate road under consideration of the HDV (gray). We compare an optimizing planner and an RL-based planner for deployment on an edge server.

Connectivity between vehicles can also be used for the coordination of maneuvers to increase traffic efficiency and safety. In the current work, we integrate and evaluate two cooperative crossing maneuver planners, as illustrated in Fig. 1. The cooperative maneuvers are planned on a centralized edge server, which instructs CAVs to explicitly deviate from priority rules to reach peak efficiency. As urban automotive traffic will not be fully automated in the near future, mixed traffic, i.e., the simultaneous use of roads by both human-driven vehicles (HDVs) and CAVs, will be prevalent. Therefore, any real-world cooperative planner must consider HDVs that cannot be directly influenced and behave according to the priority rules. We compare two approaches to cooperative maneuver planning, one of which leverages a multi-scenario prediction to derive an optimal maneuver, based on [26]. The second approach employs reinforcement learning (RL) to train a graph neural network (GNN) policy for cooperative maneuver planning [20].

Our methods are designed for urban intersections with one lane per crossing direction. Larger intersections are beyond the scope of this paper, as they are typically managed by traffic lights and require a different cooperative planning paradigm.

To achieve realistic and comparable quantitative results, we employ the real-time simulation framework DeepSIL [37] with a state-of-the-art human driver model for HDVs. We extended the simulation framework to support multiple CAVs, maneuver

图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

during training. Our RL-based cooperative planning model has been extended to mixed traffic in [20]. This approach also does not rely on the turn direction of the HDVs to be known. In addition, the integration of the trained RL policy with a sampling-based motion planner was addressed in [21], demonstrating the applicability on real vehicles.

III. COOPERATIVE MANEUVER PLANNING MODULES

In this section, the commonalities and differences of the two evaluated maneuver planning modules are presented. In our proposed system architecture, all CAVs in the scene provide their current state and desired route in the form of a sparsely sampled polyline to the centralized cooperative planning module on the edge server. The planning module generates a joint maneuver on behavior level, which is in turn distributed to all CAVs.

A. Definition and Coordination of Cooperative Maneuvers

For both planning modules, the maneuvers are passed to the CAVs according to the maneuver coordination protocol proposal in [27]. Depending on the number of vehicles on conflicting paths, up to two maneuver waypoints (one at the intersection entry and one at the intersection exit) are passed to a given CAV. The CAVs need to fulfill constraints defined by the maneuver waypoint as

$$\mu = (\mathbf{p}, t_{\min}, t_{\max}, \nu^{\uparrow}, \nu^{\downarrow}). \quad (1)$$

The interval $[t_{\min}, t_{\max}]$ denotes the admitted time window for the vehicle to cross the waypoint $\mathbf{p}$. The fields $\nu^{\uparrow}$ and $\nu^{\downarrow}$ describe the preceding and following vehicle IDs, respectively. This additional information can be leveraged by vehicle-side motion planners to improve follow trajectory planning on lead vehicles [33]. Each CAV remains responsible for its own motion planning and validation whether the maneuver constraints can be safely incorporated. Note that lateral guidance is provided by lane centerlines in a common map that is available to all vehicle-side motion planners.

B. Environment Model Processing

The basis for both cooperative maneuver planning modules is the server-side EM. This EM is the result of fusing multiple data sources, like cooperative awareness messages (CAM, [8]) and collective perception messages (CPM, [9]) from connected vehicles or infrastructure perception [4]. It is considered to contain all information relevant for maneuver planning with full observation of the intersection area. A vehicle in the EM is denoted as

$$\nu_{\text{id}} = (\mathbf{T}, v, \mathcal{D}, c) \in \text{EM}, \quad (2)$$

where id denotes a unique identifier and the binary flag c determines whether the vehicle is a CAV and thus controllable. $\mathbf{T}$ describes the vehicle's pose on the local 2D ground plane and v is the current driving speed. CAVs share their intended route or destination $\mathcal{D}$ as a set of sparse waypoints, which is unknown for HDVs, though, which are assumed to be non-connected.

TABLE I
OBSERVATION INPUT FEATURES OF THE TWO PREDICTION MLPs OF THE OPTIMIZATION-BASED PLANNING MODULE. TABLE ADAPTED FROM [28].

Environment observation $\mathbf{o}_i$ of ν_i	Input features
Distance to stop line	$d_{\text{stop},i}$
Current velocity and speed limit	$v_i, v_{\text{max},i}$
Relative lane heading in n meters	$\Delta\psi_{i,-10}, \Delta\psi_{i,-3}, \Delta\psi_{i,3}, \Delta\psi_{i,10}, \Delta\psi_{i,30}, \Delta\psi_{i,100}$
Lead vehicle distance and velocity	$d_{\text{lead},i}, v_{\text{lead},i}$
Gap obs. $\mathbf{o}_{i,j}$ of ν_i towards ν_j	Input features
Distance to end of conflict zone	$d_{\text{target},i}$
Velocity of this and the other vehicle	v_i, v_j
Distance of other vehicle to stop line	$d_{\text{stop},j}$

In a preprocessing step, both planning modules associate and project the vehicles in the server-side EM to the lane centerlines of an internal map format. While connected vehicles share their routes with the planning module, the turn intention of human drivers is unknown. Thus, both planners conservatively assume that an HDV may be in conflict with any crossing or merging lanes. Additionally, it is assumed that HDVs do not know the route of other vehicles and that the indicator light states cannot be reliably evaluated. The handling of this uncertainty might be improved by employing a prediction algorithm such as [36], which is out of scope for this work, though. Moreover, vulnerable road users that are not lane-bound are not accounted for in the present setup.

C. Optimization-based Planning Algorithm

The optimization-based planner builds upon our previous works presented in [26]. The main component is a scene-consistent prediction module that estimates the behavior of the vehicles of the current traffic scene in the immediate future. Multiple such scenario predictions with various applicable maneuvers are performed and the best one—according to the validity and efficiency metrics defined below—is selected.

1) *Driver Model for Prediction:* Vital to the maneuver planning is the scenario prediction module that we previously published in [28]. We employ a longitudinal lane-based prediction module trained on real-world traffic data. The driver model comprises two multi-layer perceptrons (MLPs) with 2 hidden layers of size 16 each: MLP_{acc} estimates the acceleration during the next time step. MLP_{gap} determines the gap acceptance, i.e., the decision whether a side road vehicle should merge or cross in a gap in main road traffic. The driver model is evaluated jointly for all vehicles in the traffic scene and integrated over a 15 s horizon with a time step of 0.1 s. This prediction is run for each priority assignment as described in Section III-C2. The input features are shown in Table I and Fig. 2.

The acceleration estimation MLP_{acc} was trained using proximal policy optimization (PPO) [35] in an RL closed-loop simulation environment on different intersection layouts. The objective function encourages driving near the speed limit and penalizes collisions with prioritized vehicles to emulate natural driving behavior. The gap acceptance model MLP_{gap}

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.3, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

The first part of the network architecture is comprised of MLPs that encode the vertex input features and edge input features, respectively. At the core of the GNN, modified relational graph convolutional network layers [34] and graph attention layers [38] are used for message passing in an interleaved structure. In the actor network architecture, the resulting vertex features are mapped to a joint action by a final output MLP. In contrast, the critic network aggregates the vertex feature vectors resulting from message passing to a single feature vector before decoding a Q-value estimate using another MLP. The graph representation and the GNN are implemented using the PyTorch Geometric API [10].

Apart from a suited network architecture, RL requires a well-designed reward function to learn a reasonable policy. Details on the definition and tuning of the various reward components can be found in [19], [20].

RL planning runs are triggered regularly With a period of 2 s, which facilitates the observation of the CAVs' reactions to the cooperative maneuver with the start of the next planning run. For typical traffic scenarios at smaller intersections, the planning run takes significantly less than the period length of 2 s. In unfavorable circumstances with many (non-connected) vehicles, a full planning run might take longer, which is caught by a timeout that ensures real-time capability [21].

E. Prediction and Derivation of Cooperative Maneuvers

For the optimization-based planner, the selected priority assignments need to be converted to maneuver constraints as in Eq. (1) for each individual CAV. Each pair of conflicting CAVs shares a conflict zone, i.e., a segment along the road that may only be occupied by one of the vehicles at any time to guarantee freedom from collision, as explained above. The predicted timestamp at which the prioritized vehicle is predicted to leave the conflict zone is used to encode the space-time maneuver constraints for entering and exiting. The start and end points of the conflict zone along the respective vehicle routes are used as waypoints for the constraints.

The RL-based cooperative planner derives cooperative maneuvers using a built-in simulator. After initializing the simulator based on the current state of the server-side EM, the RL policy is queried to predict the future evolution of the scene. Analogously to the optimization-based planner, the crossing points of the simulated trajectory and the intersection conflict zone are used to derive so-called waypoint candidates. Only if there is a conflicting vehicle, the corresponding candidate waypoint is used as a maneuver waypoint. Thus, only if there is a preceding vehicle, the waypoint at the intersection entry is included in the maneuver. Otherwise, the CAV is free to enter the intersection as soon as it considers it beneficial to the maneuver, which leaves room for optimization to the CAV motion planning.

As noted above, the CAVs remain responsible to check the maneuver for local feasibility, and the maneuver planners cannot guarantee that the CAVs can execute the suggested maneuver. However, while the high-level maneuver planning is optimized for speed, the driver models internally used in each maneuver planner strive to resemble realistic behavior,

which usually is far from the physical (or even comfortable) limits at intersections. Thus, the coordinated CAVs can usually easily comply with the required time constraints.

IV. SIMULATION APPROACH

We integrated the proposed cooperative planning approaches into an extended simulation and evaluation framework for fully automated and mixed traffic. The cooperative planners are evaluated and assessed against two baselines. Our primary baseline are non-cooperative (NC) CAVs that leverage shared perception data as in [4], but no routing information and adhere to priority rules. This baseline employs the same underlying trajectory planning as is used for the cooperative approaches. Thus, it constitutes a rather strong baseline, which facilitates a sharp evaluation of the cooperative aspect in isolation. In addition, we present the metrics for pure HDV traffic, which is simulated using a learned behavior model. As such, it resembles human driving behavior more closely and exhibits a human-typical agile behavior. The focus of this paper is on the comparison of the two cooperative planning approaches at one exemplary intersection. We refer the reader to previous publications for more extensive individual evaluations of the optimization-based [26] and the RL-based [20] cooperative planners, respectively.

A. Simulation Setup

The simulation framework that we used for evaluation of the planning methods builds upon the DeepSIL simulator introduced in [37]. It is based on the ROS2 software distribution [25] and can simulate HDVs as well as CAVs. We extended the framework to support multiple CAVs and cooperative maneuvers. The simulation has a base time step of 50 ms and runs in wall-clock time. This means that processing delays from the cooperative maneuver planners are considered and affect the results, while communication delays are not modeled.

We use a single potent simulation machine with an AMD Ryzen 3990X CPU. Still, running multiple instances of large parts of the CAV software stack as well as multiple deep-learning-based HDV models requires a lot of computational resources, while the maneuver planning requirements are comparatively modest. Due to those limitations, the simulations need to be limited to eight vehicles total. Additionally, scenarios with more than four CAVs are slowed down slightly to cope with immaturity in the ROS2 communication layer.

Human-driven vehicles are predicted using the approach in [37]. Every 50 ms, the multiple trajectory prediction network is evaluated on the current traffic environment. In each cycle, the most progressing, non-conflicting prediction is used to update the vehicle position and velocity. The prediction is converted into a longitudinal action and the internal vehicle simulation is stepped forward along the centerline of the route. A simple occlusion simulation was added, i.e., HDVs on the subordinate road approach slowly until they reach a distance of 2 m to the intersection, at which a full view into the situation is assumed. Also, the simulated HDVs do not know the routes

图 3: 核心方法或算法页 / Core method or algorithm page (p.5, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

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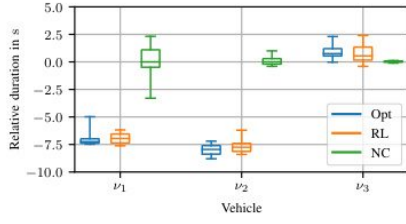


Fig. 8. Real-world maneuver durations for each involved CAV relative to the median baseline duration.

conflict with ν_1 but only with ν_3 . Fig. 1 illustrates a typical cooperative maneuver. If vehicle ν_3 waives its right of way and possibly slows down while approaching, it allows both other CAVs to cross the intersection virtually simultaneously. Depending on the timing, vehicles ν_1 and ν_2 can simply traverse the intersection unimpeded. Ideally, vehicle ν_3 should not be required to stop, but slow down smoothly and accelerate again once the intersection is cleared.

B. Evaluation in Real-World

To obtain a reliable quantitative evaluation in real traffic, the initial scenario depicted in Fig. 1 was driven numerous times. Thereof, five baseline runs have been performed by driving according to priority rules, twelve runs under the command of the optimization-based planner, and 20 runs employing the RL planner. The cooperative maneuvers have been planned reactively with respect to surrounding traffic, while the CAVs approach the intersection. This manifests in alternative maneuvers being proposed if the original proposal turns out to be less effective or even infeasible. One of the evaluation runs using the RL planner was recorded on video for demonstration purposes and is available at [32]. The vehicle trajectories have been recorded and clipped to the common evaluation segment, which was also used for the simulative evaluation. Because all experiments have been conducted in real traffic at a public intersection, each run had slightly different initial conditions. While the safety drivers guided the test vehicles carefully to the engagement point, minor deviations in timing and initial speed remain. During all maneuver runs that are contained in the evaluation, the automated driving system was successfully engaged.

One of the most expressive metrics for traffic efficiency at intersections is the delay induced by interaction with other vehicles. In the present work, we consider the maneuver duration, i.e., the time it takes for a vehicle to cross the evaluation segment, relative to the median baseline duration. Therefore, the median bars of the baseline distribution in the boxplot in Fig. 8 are exactly at a duration of 0 s. The outlier of vehicle ν_1 at around -3.5 s can be explained by a deviation in initial timing. In this run, the vehicle did not have to stop and its driving speed remained above $3 \frac{\text{m}}{\text{s}}$. It can be observed that both cooperative planning approaches yield a significant reduction in maneuver duration for the vehicles ν_1

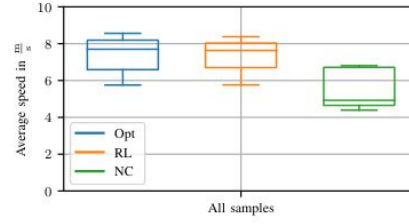


Fig. 9. Real-world driving speed averaged over all CAVs when coordinated by one of the cooperative planners compared to baseline driving.

and ν_2 , which cross the intersection simultaneously in the cooperative maneuver. While the time gain for each of these vehicles is roughly 7 s, the vehicle ν_3 on the major road rarely loses more than 2 s during the approach and traversal of the intersection. It can be concluded that both cooperative planners are able to identify traffic scenes in which the ordinary priority rules yield sub-optimal flow and propose a more efficient maneuver.

In addition, we present the average speed distribution over all vehicles in Fig. 9. Both cooperative planners yield a mean driving speed of around $7.5 \frac{\text{m}}{\text{s}}$ ($27 \frac{\text{km}}{\text{h}}$) over the evaluated trajectory segment. Considering the lane curvature on the confined intersection space, this indicates a swift intersection traversal without any stops. In contrast, when driving according to priority rules, the average speed remains below $5 \frac{\text{m}}{\text{s}}$. Undoubtedly, the yielding behavior of the vehicles ν_1 and ν_2 might lead to stops in this case, which causes a decrease in driving speed and efficiency.

The three real-world CAVs all use different motion planning algorithms that were adapted to demonstrate the cooperative use case and were not yet fully tuned to exactly meet the time constraints resulting from the maneuver planning. Therefore, a criticality metric evaluation on the real-world test drives would not be of high significance, which should, of course, be addressed by further research.

VII. CONCLUSION

In this paper, an optimization-based and an RL-based approach for cooperative maneuver planning in mixed traffic at urban intersections have been evaluated in a realistic simulation and in real-world experiments. Especially in simulated fully automated traffic, cooperative maneuvers achieved a significant improvement of traffic efficiency. Even in simulated mixed traffic with only 50 % CAV penetration, the planning approaches successfully performed cooperative maneuvers in about one third of the scenarios and increased the efficiency, despite not relying on any cooperation from HDVs. Overall, the optimization-based planner yielded higher efficiency gains, while also causing a larger increase of criticality measures. Experiments with three prototype CAVs in public traffic demonstrated the real-world applicability of both cooperative planners and showed a significant efficiency increase. Future work will include the evaluation of further cooperative use

图 4: 结论或局限页 / Conclusion or limitations page (p.9, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

planning modules, and V2X communication in between. Also, we integrated the trajectory planner and controller combination from [33] as well as an advanced vehicle model from one of our real-world CAVs into the simulation. Additionally, to the best of our knowledge, we are the first to provide an experimental evaluation of cooperative planners in public mixed traffic using real test vehicles and V2X communication. The execution of the cooperative maneuvers in simulation and in real-world is based on the coordination protocol proposed in [27]. Thus, the core contribution of the present work is threefold:

- Extension of our simulation framework to integrate cooperative maneuvers and realistic HDV and CAV models;
- Statistical analysis and evaluation of our two planning approaches in fully automated and in mixed traffic;
- Demonstration of the real-world applicability of both planners using three prototypical CAVs in public traffic.

The remainder of the paper is structured as follows: Section II discusses the state of the art in classical and learning-based cooperative planning for automated driving. Our previously introduced maneuver planning approaches are summarized and compared in Section III. We give an in-depth overview over our simulation framework and evaluation method in Section IV. Afterwards, we discuss our simulative evaluation results (Section V) and real-world experiments in public traffic (Section VI). Finally, Section VII summarizes the results of the article and gives an outlook on future work.

II. RELATED WORK

The efficient coordination of CAVs at urban intersections receives strong research interest in the field of automatic intersection management (AIM). In the following, we give an overview on published works that are deemed most relevant to cooperative maneuver planning at unsignalized intersections. Prior works mostly consider fully automated traffic and pure simulative evaluations, as surveyed by [46]. Also, the evaluation of the approaches often does not consider the processing time, and many researchers do not consider a communication protocol to coordinate the traffic. Therefore, most previous works are not suitable for real-world deployment.

A. Classical Cooperative Maneuver Planning

The earliest cooperative planning methods rely on an exhaustive vehicle ordering search [24] or a simple heuristic such as first-come, first-served [6], [42]. Later approaches try to find a near-optimal crossing order of the arriving vehicles using more sophisticated algorithms like dynamic programming [43], ant colonies [40], a control policy based on Petri Nets [1], Mixed Integer Quadratic Programs [15], or Monte Carlo Tree Search [22]. However, all of these works are designed for CAV-only traffic and most were evaluated in simplistic simulation environments. Only the authors in [1] show the real-time capability of their algorithm in simulation and in real-world experiments.

Fewer researchers have tackled the intersection management task in the much more challenging mixed traffic setting and especially scenarios with a vast majority of HDVs, which

will be prevalent in the near future. To the best of our knowledge, all previous mixed traffic approaches need to incorporate traffic lights to support the presence of HDVs. The first notable approaches [2], [7] were designed for a low rate of HDVs ($< 10\%$) and the gains deteriorate in an environment with a higher amount of HDVs. Two later publications [30], [44] proposed virtual platoons for signalized intersection management, supporting mixed traffic at high HDV rates and being evaluated in a real-time traffic simulation. However, all methods mentioned above require traffic lights to control HDVs and as such are out of our problem scope. Also, most approaches assume that the turn direction of HDVs is known, which is generally not given in real-world traffic. Prior works claiming HDV support without traffic lights or other means of communication with human drivers depend on strong assumptions on the HDVs' behavior in mixed traffic. The reservation-based approach [45] assigns CAVs a priority based on their lane, while expecting HDVs to follow a possibly present leading CAV closely. The application of such an approach in public traffic where HDVs are not specifically instructed remains questionable.

Our optimization-based approach proposed in [26] is specifically designed to handle unsignalized intersections with any HDV rate and unknown HDV turn directions. It employs a scene-consistent multi-scenario prediction and generates maneuvers between the present CAVs to optimize the expected efficiency. A respective coordination protocol to execute the cooperative maneuvers on CAVs was previously proposed in [27], making this approach applicable on real vehicles.

B. Learning-based Cooperative Maneuver Planning

Although non-learning approaches are still prevalent in AIM, machine learning is also gaining traction in connected automated driving. In [5], it is proposed to leverage collective perception to improve the performance of local planning algorithms. The planning task only considers one automated vehicle without cooperative objective, though. Due to lack of ground-truth data for training, learning-based approaches to AIM typically rely on RL. Such a learning-based approach is proposed by [41] that suggests to train a policy through RL to choose from a restricted action space that ensures collision-free maneuvers. However, this approach is designed for CAV-only traffic and the evaluation relies solely on simulations.

In [31], the authors propose an RL approach managing intersection traversals in presence of simulated human-driven vehicles while turning maneuvers are disallowed. A further RL framework that combines local observations with a joint reward to accommodate a cooperative objective was presented by [12] and is capable of handling mixed traffic, though with a limited number of cooperating vehicles in the scene. Both works lack the application to real vehicles and a comprehensive evaluation.

In our previous works [18], [19], we have proposed a flexible graph-based scene representation and an RL training scheme for AIM in fully automated traffic. It was shown that the model outperforms a FIFO baseline and generalizes within certain limits to intersection layouts not encountered

图 5: 补充精读页 / Supplemental close-reading page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Real-world cooperative intersection management：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.2 2 REAL-WORLD EVALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 论文不是提出单一新控制算法，而是比较多场景预测和图强化学习等协同机动规划模块，在混合交通仿真与真实车辆中评估停车、时间和关键性。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.2 II. RELATEDWORK

章节目标 / Learning goal: 建立方法谱系和相关工作边界，避免把本文贡献误读为完全从零开始。

Place the paper inside the broader method taxonomy.

关键论点 / Key argument: 该部分说明作者站在哪些已有工作之上，以及本文选择绕开或继承了哪些限制。

技术焦点 / Technical focus: 把相关工作按 scheduler、smoother、safety filter、evaluation 四类重排。

审稿追问 / Reviewer question: 作者是否公平比较了最接近的强 baseline？

p.3 III. COOPERATIVE MANEUVER PLANNING MODULES

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.4 4 REAL-WORLD EVALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 论文不是提出单一新控制算法，而是比较多场景预测和图强化学习等协同机动规划模块，在混合交通仿真与真实车辆中评估停车、时间和关键性。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.5 IV. SIMULATION APPROACH

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

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审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.6 6 REAL-WORLD EVALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

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审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.6 V. SIMULATION EVALUATION RESULTS

章节目标 / Learning goal: 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环: Novel mixed-traffic simulation framework plus real-world prototype connected automated vehicle drives.

技术焦点 / Technical focus: 重点追踪指标: 穿越时间、停车次数、关键性/风险、成功率和真实系统运行稳定性。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益, 还是覆盖了高密度、异常扰动和失败案例?

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题: 协同交叉口管理方法在混合交通仿真与真实原型车测试中是否仍能减少停车和通行时间? 读这一段时不要急着接受动机, 要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括: 原型系统中的通信与感知可支持协同机动规划。; 真实测试场景代表性足以支持初步结论。; 关键性指标可近似反映真实安全风险。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“评估指标聚合 / Evaluation-metric aggregation”复写符号表, 确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流: 输入交通状态, 执行 Compares multi-scenario prediction and graph-based reinforcement learning approaches in mixed-traffic simulation and real drives., 输出可执行通行/控制决策, 并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升, 要检查基准、公平性和指标。本文验证描述为: Novel mixed-traffic simulation framework plus real-world prototype connected automated vehicle drives.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是: Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	评估指标聚合 / Evaluation-metric aggregation
3	核心求解 / Core solver	Compares multi-scenario prediction and graph-based reinforcement learning approaches in mixed-traffic simulation and real drives.
4	安全接口 / Safety interface	Evaluates criticality metrics while operating with prototype connected automated vehicles in public traffic.
5	平滑与效率 / Smoothing and efficiency	Reports substantial reductions in crossing time and number of stops for cooperative maneuver planning.
6	验证 / Validation	Novel mixed-traffic simulation framework plus real-world prototype connected automated vehicle drives.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	3/5	评估指标聚合 / Evaluation-metric aggregation
算法贡献 / Algorithmic contribution	5/5	Compares multi-scenario prediction and graph-based reinforcement learning approaches in mixed-traffic simulation and real drives.
实验贡献 / Experimental contribution	5/5	Novel mixed-traffic simulation framework plus real-world prototype connected automated vehicle drives.
工程贡献 / Engineering contribution	5/5	Evaluates criticality metrics while operating with prototype connected automated vehicles in public traffic.
博士课题价值 / PhD relevance	4/5	Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

评估指标聚合 / Evaluation-metric aggregation

$$S = \frac{1}{M} \sum_{m=1}^M n_m^{stop}, \quad T = \frac{1}{M} \sum_{m=1}^M (t_m^{out} - t_m^{in}), \quad K = \max_m \kappa_m$$

Symbol / 符号	含义	Meaning	Role / 建模作用
S	平均停车次数	average number of stops	衡量 stop avoidance 的直接指标
T	平均穿越时间	average crossing time	衡量效率
K	最大关键性	maximum criticality	表示最危险场景风险
M	评估运行次数	number of evaluation runs	仿真或真实试验样本量
kappa_m	关键性指标	criticality measure	安全评估变量

5.2 核心算法架构 / Core Algorithm Layout

论文不是提出单一新控制算法，而是比较多场景预测和图强化学习等协同机动规划模块，在混合交通仿真与真实车辆中评估停车、时间和关键性。

Rather than only proposing a controller, the paper evaluates cooperative maneuver-planning modules in simulation and real-world prototype drives.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

贡献在验证闭环：真实车辆实验暴露通信、定位、感知和人类交通干扰。它为仿真论文提供“哪些指标必须走向真实”的标尺。

Its value is validation realism: real drives reveal communication, localization, perception, and mixed-traffic issues hidden in pure simulation.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 原型系统中的通信与感知可支持协同机动规划。
- 真实测试场景代表性足以支持初步结论。
- 关键性指标可近似反映真实安全风险。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：不同协同机动规划模块、非协同驾驶、仿真基线 and 真实驾驶对照。

Validation / 验证： Novel mixed-traffic simulation framework plus real-world prototype connected automated vehicle drives.

Metrics / 指标： 穿越时间、停车次数、关键性/风险、成功率和真实系统运行稳定性。

Ablation / 消融： 作为评估论文，模块消融不一定充分；最需要的是同一交通需求下的模块替换、传感器/通信降级和 HDV 行为差异测试。

Reviewer reading / 审稿式阅读提醒： 阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 33 figure references and 4 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
策略网络/训练流程图	policy-network or training-pipeline diagram	检查状态、动作、奖励、经验回放和安全惩罚是否闭环一致。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 样本规模和场景覆盖通常小于大规模仿真。
- 不是 JSSP 建模论文，理论调度贡献有限。
- 真实测试条件可能较温和，极端场景不足。

7.2 博士课题拓展点 / Research Extensions for PhD

- 把你的 JSSP scheduler 输出转化为相同停车/穿越时间/关键性指标，形成可对标实验。
- 设计 sim-to-real gap 表格：仿真中通过的策略在真实系统中哪些环节会失效。
- 加入通信延迟、定位噪声和 HDV 插入，复现实验鲁棒性。

7.3 如何服务当前课题 / Use in This Project

Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 通信延迟 / Communication latency, range [0, 500] ms.

延迟上升会降低协同收益，并增加保守安全缓冲。

Higher latency reduces coordination benefit and increases safety buffers.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator, 右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block, 紧跟符号表。
- 审稿人批注用 reviewer-note block, 博士拓展用 research-idea list。
- 交互参数用 `input[type=range]` + canvas 曲线, 打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION
p.2	2 REAL-WORLD EV ALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES
p.5	IV. SIMULATION APPROACH

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 2 REAL-WORLD EV ALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.2 II. RELATEDWORK 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.3 III. COOPERATIVEMANEUVERPLANNINGMODULES 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.4 4 REAL-WORLD EV ALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.5 IV. SIMULATION APPROACH 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。 / Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.6 6 REAL-WORLD EV ALUATION OF TWO COOPERATIVE INTERSECTION MANAGEMENT APPROACHES 作为局部论证段落阅读，关注它如何服务主问题。 / Read as a local argument and ask how it supports the main question.

p.6 V. SIMULATION EVALUATION RESULTS 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。 / Tests efficiency, safety, or energy claims; inspect whether baselines are fair.